Waits and Queues and You

Thomas LaRock
Senior DBA, Confio Software

Ignite IT Performance™

Who Am I?

Microsoft CERTIFIED Database Administrator

Microsoft MVP Most Valuable Professional

PASS

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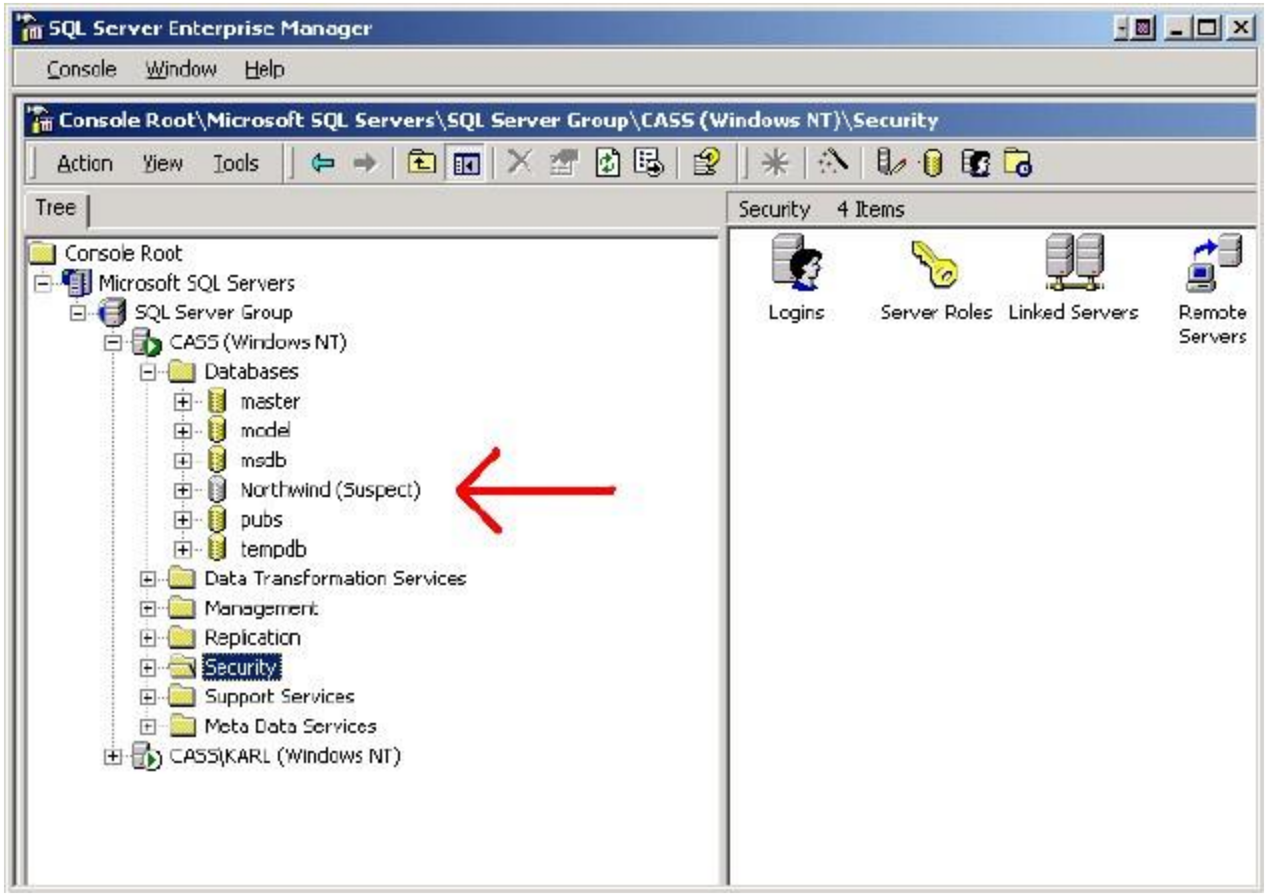
@SQLRockstar

http://thomaslarock.com

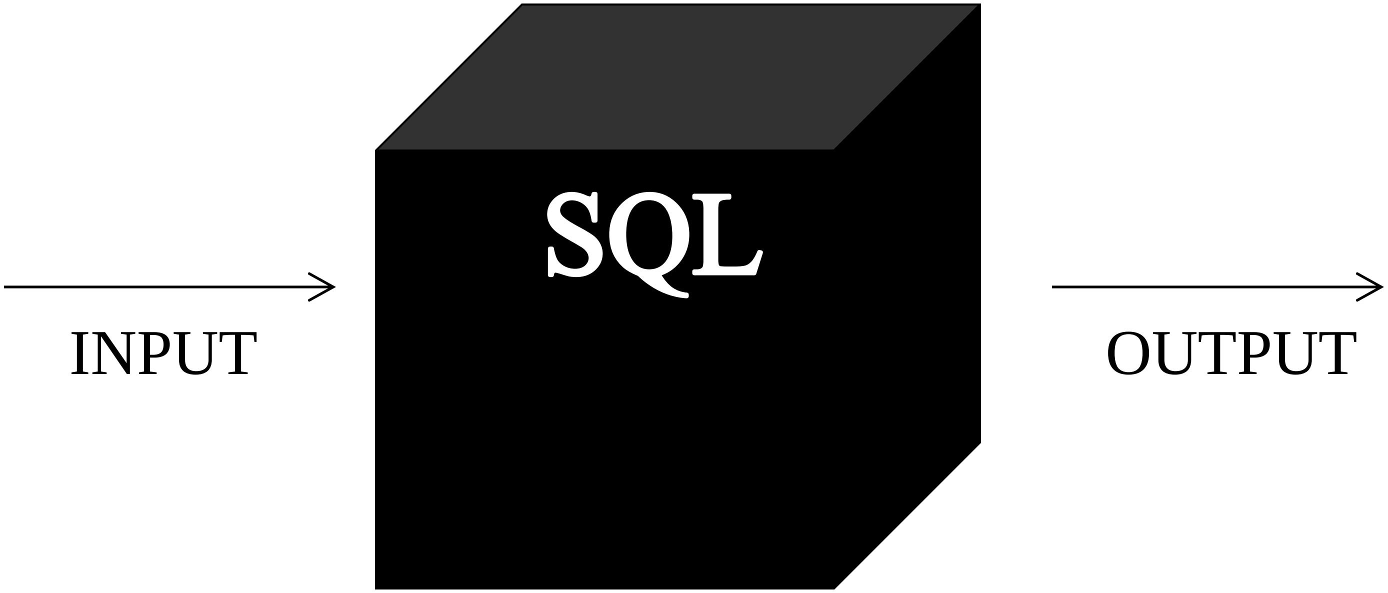
Learning to Drive

Learning to DBA



Car Engine

SQL Server is a Black Box



SQLOS

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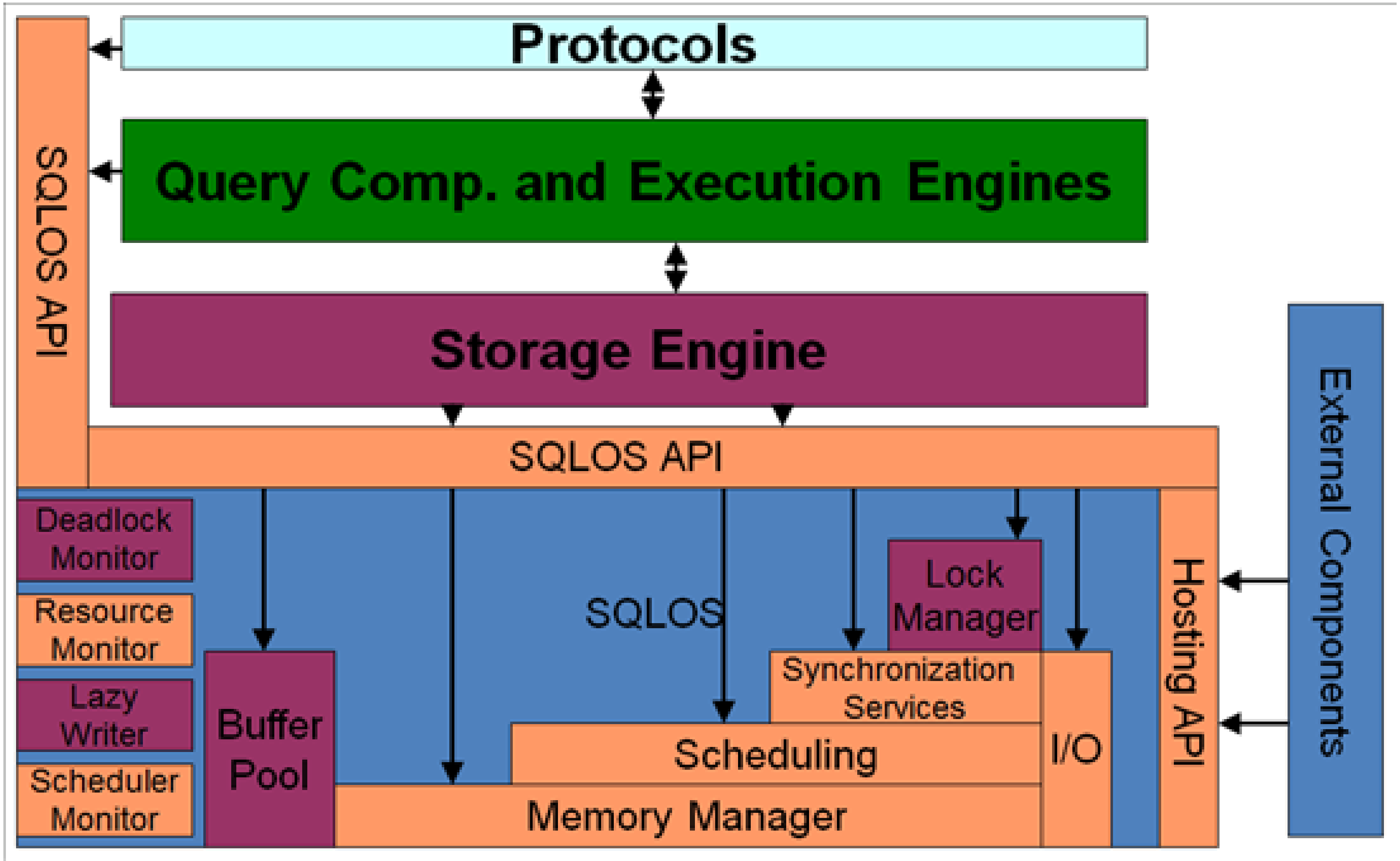


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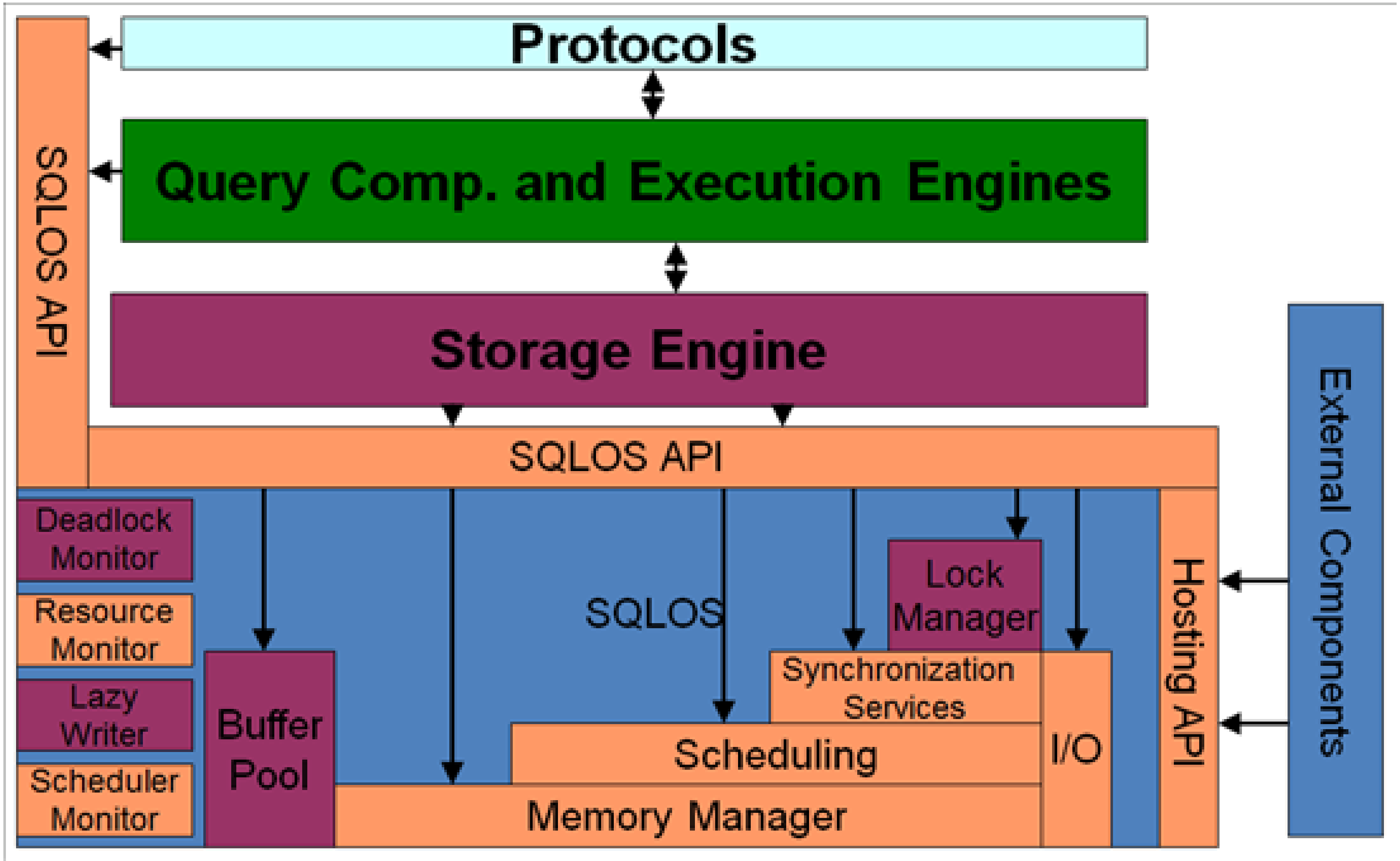
SQLOS


Protocols
Query Comp. and Execution Engines
Storage Engine
SQLOS API
SQLOS API
Deadlock Monitor
Resource Monitor
Lazy Writer
Scheduler Monitor
Buffer Pool
SQLOS
Lock Manager
Synchronization Services
Scheduling
Memory Manager
I/O
Hosting API
External Components

Agenda

- Describe Execution Model

Agenda

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- Use some DMVs

Agenda

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- Describe Execution Model
- Use some DMVs
- Show why we measure waits (use case)

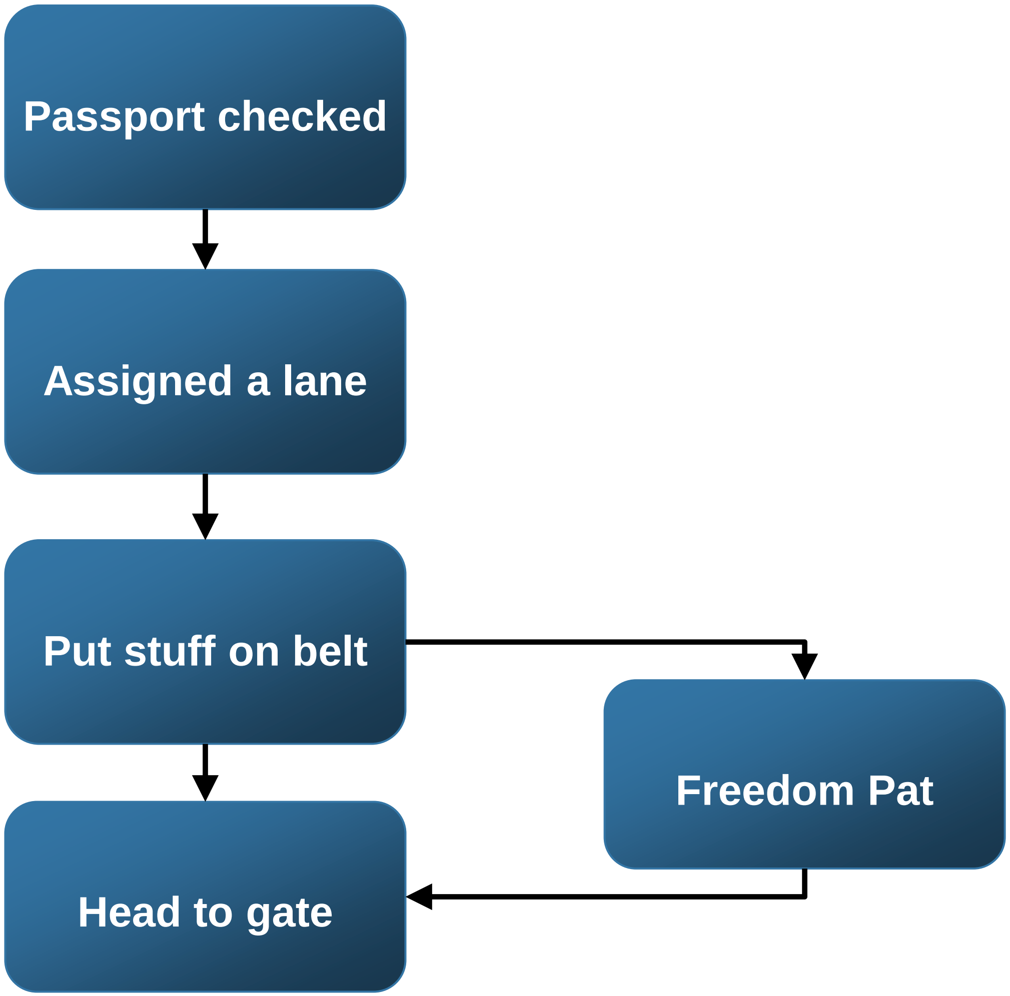
Execution Model Example

Your Tax Dollars at Work

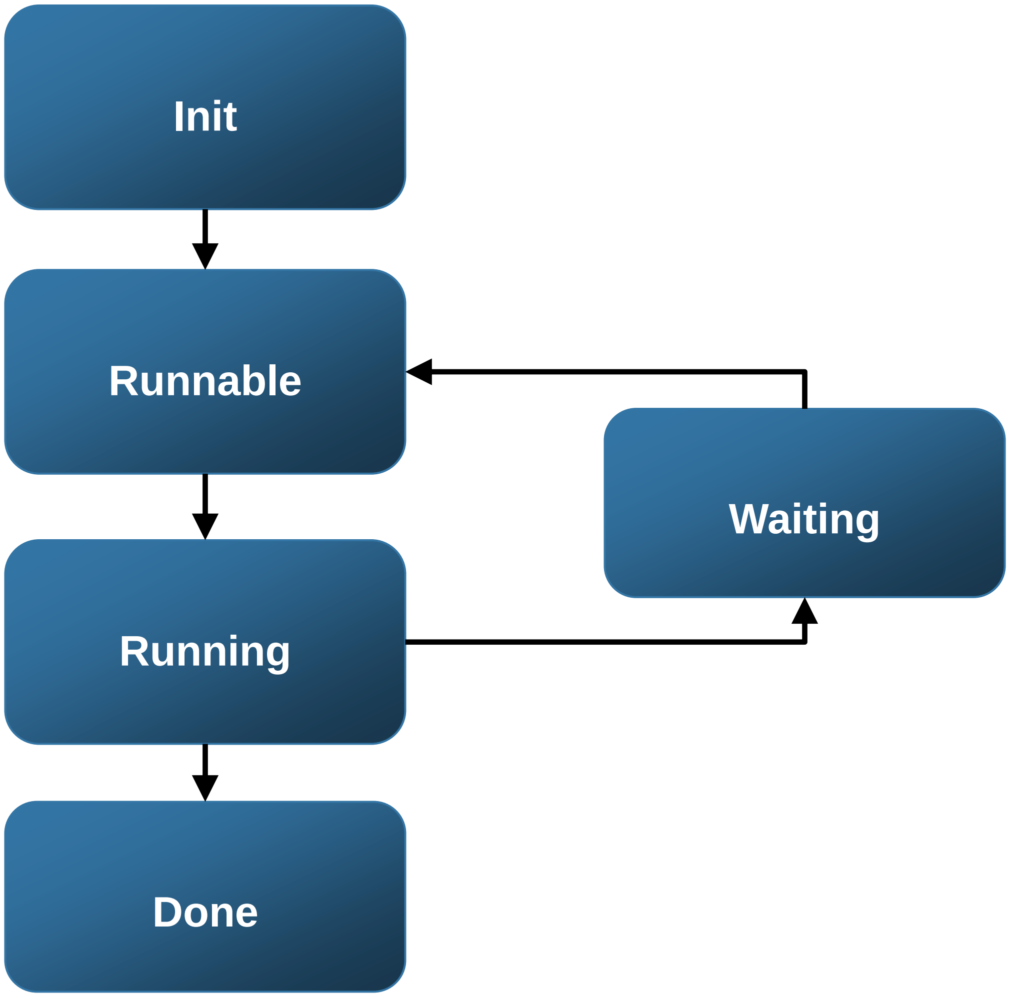
Your Tax Dollars at Work


EYEWITNESS NEWS | HD

Execution Model (TSA)



Execution Model (SQLOS)



Execution Model (SQLOS)

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CPU 1

SPID 77 – Running

CPU 1 Queue

SPID 81 – Runnable
SPID 91 – Runnable

Waiter Queue

SPID 62 – ASYNC_NETWORK_IO
SPID 63 – CXPACKET
SPID 67 – PAGEIOLATCH_SH
SPID 74 – LCK_M_S
SPID 79 – LATCH_EX
SPID 87 – WRITELOG

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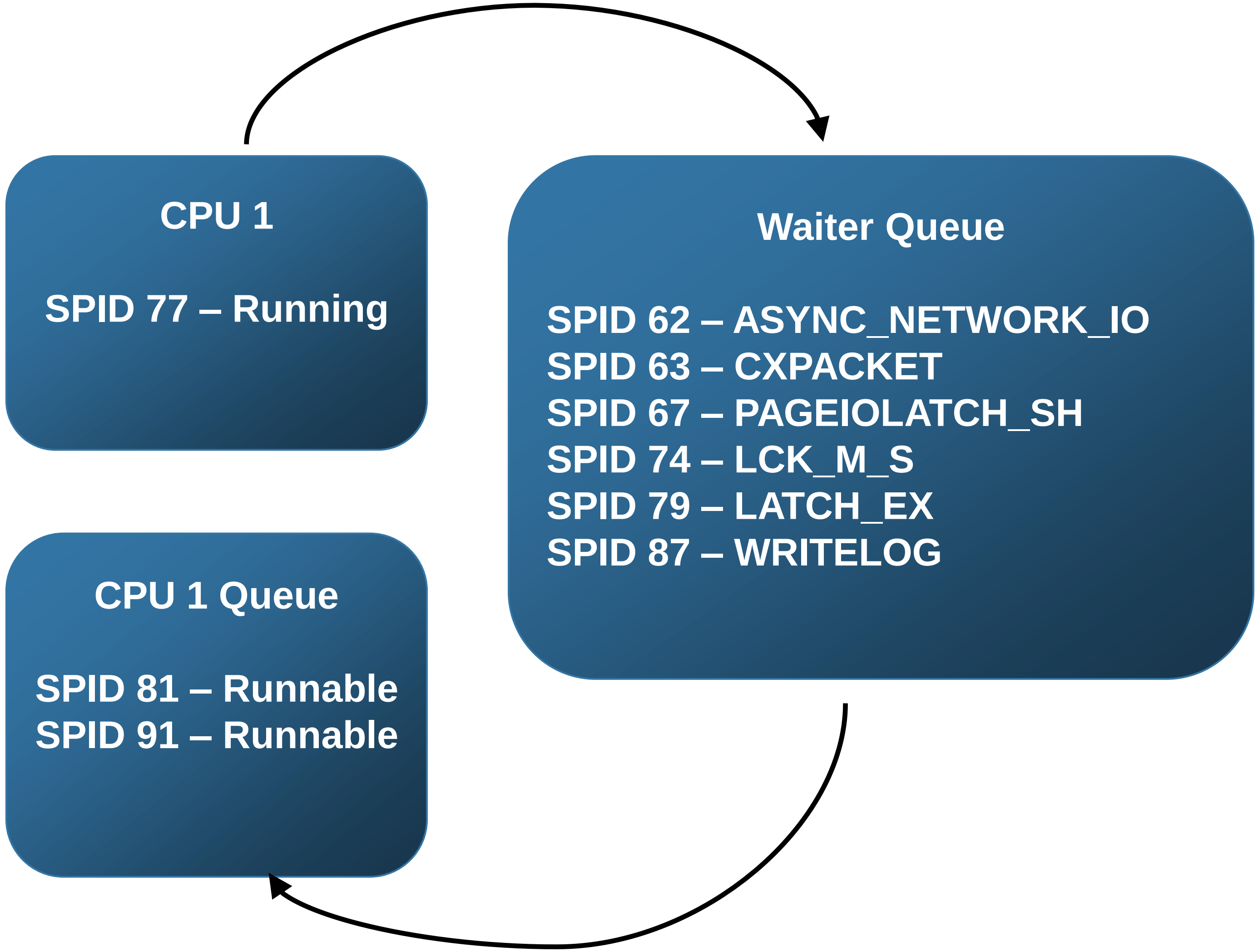

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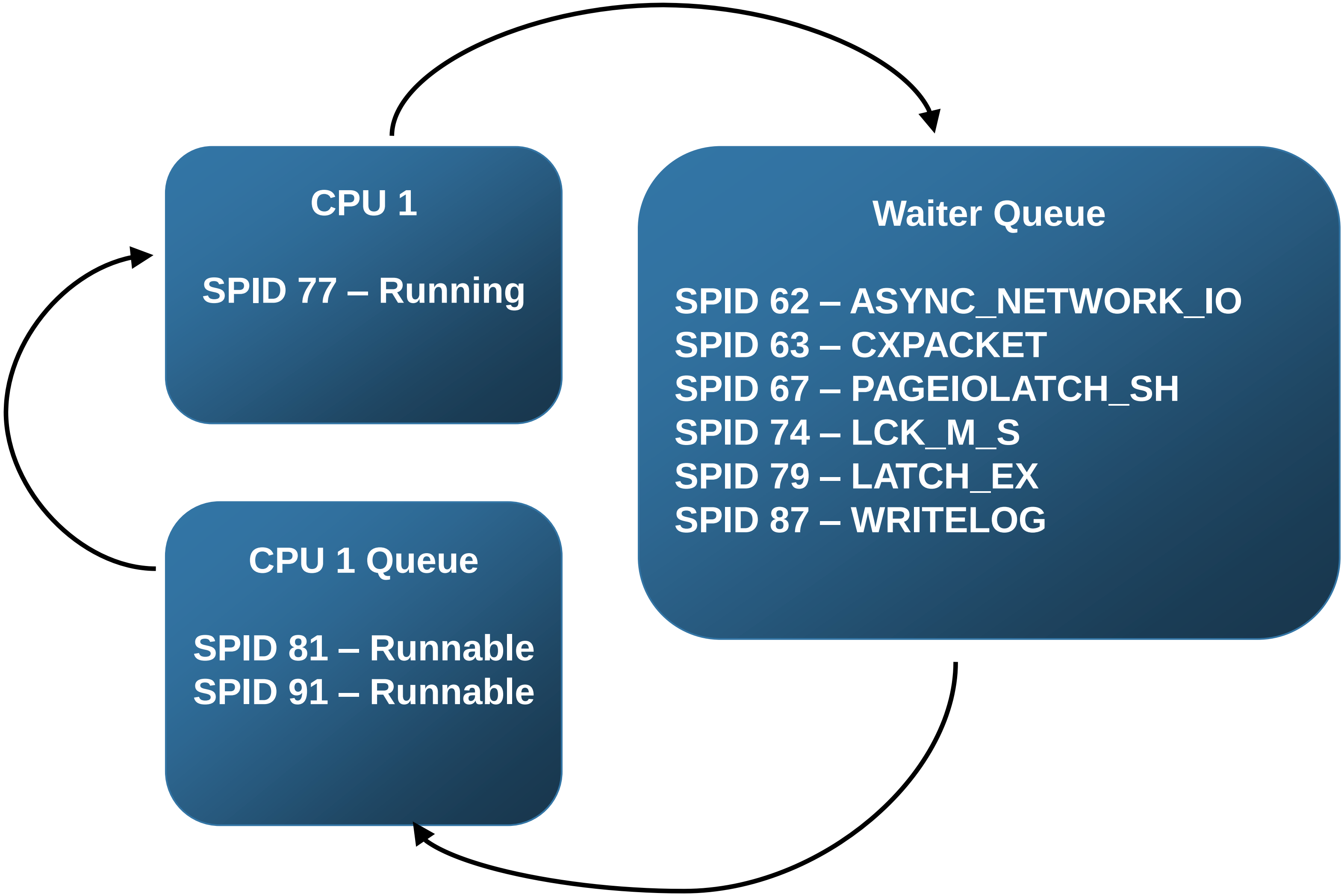
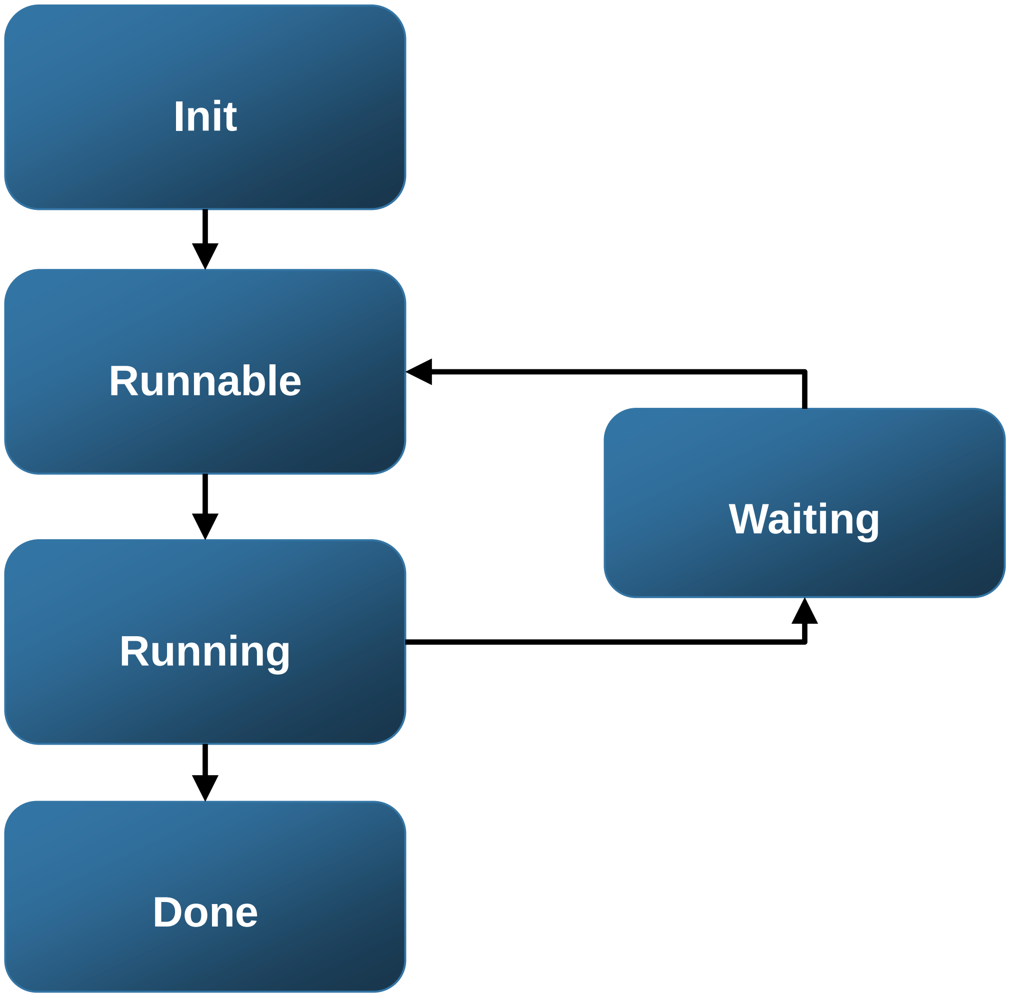


Execution Model (DMVs)



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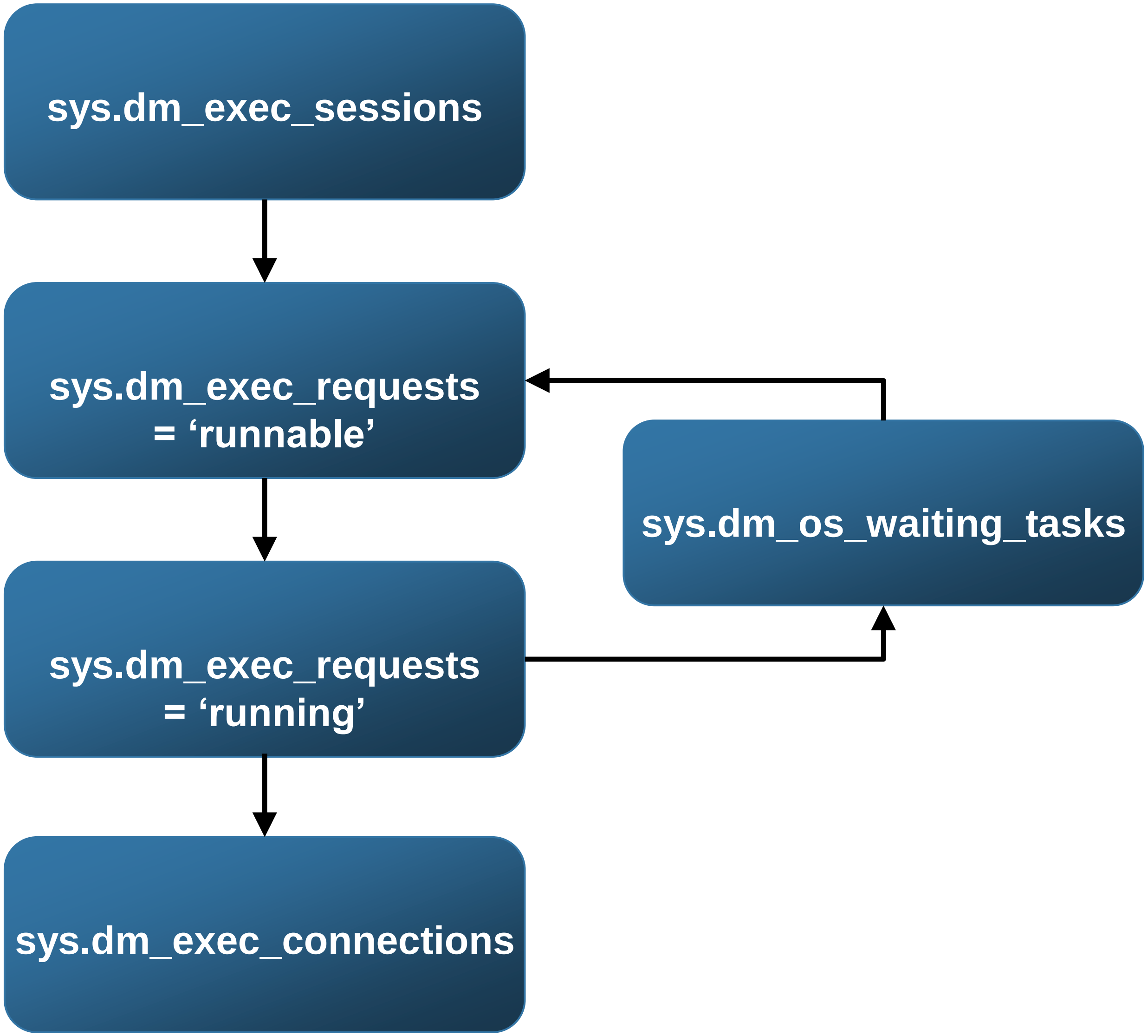

Demo

DEMO

Sample Wait Types

1.

2.

3.

4.

5.

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5. LCK_M_S, LCK_M_U, LCK_M_X…

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Sample Wait Types

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   - Waiting for a log flush to complete
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    - Parallelism
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Sample Wait Types

1. ASYNC_NETWORK_IO, NETWORKIO
   - Waiting on the network
2. CXPACKET
   - Parallelism
3. WRITELOG
   - Waiting for a log flush to complete
4. PAGEIOLATCH_SH, PAGEIOLATCH_EX…
   - Physical disk reads
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Why Measure Waits?

- Think about how you currently resolve performance issues, does it look like this…?

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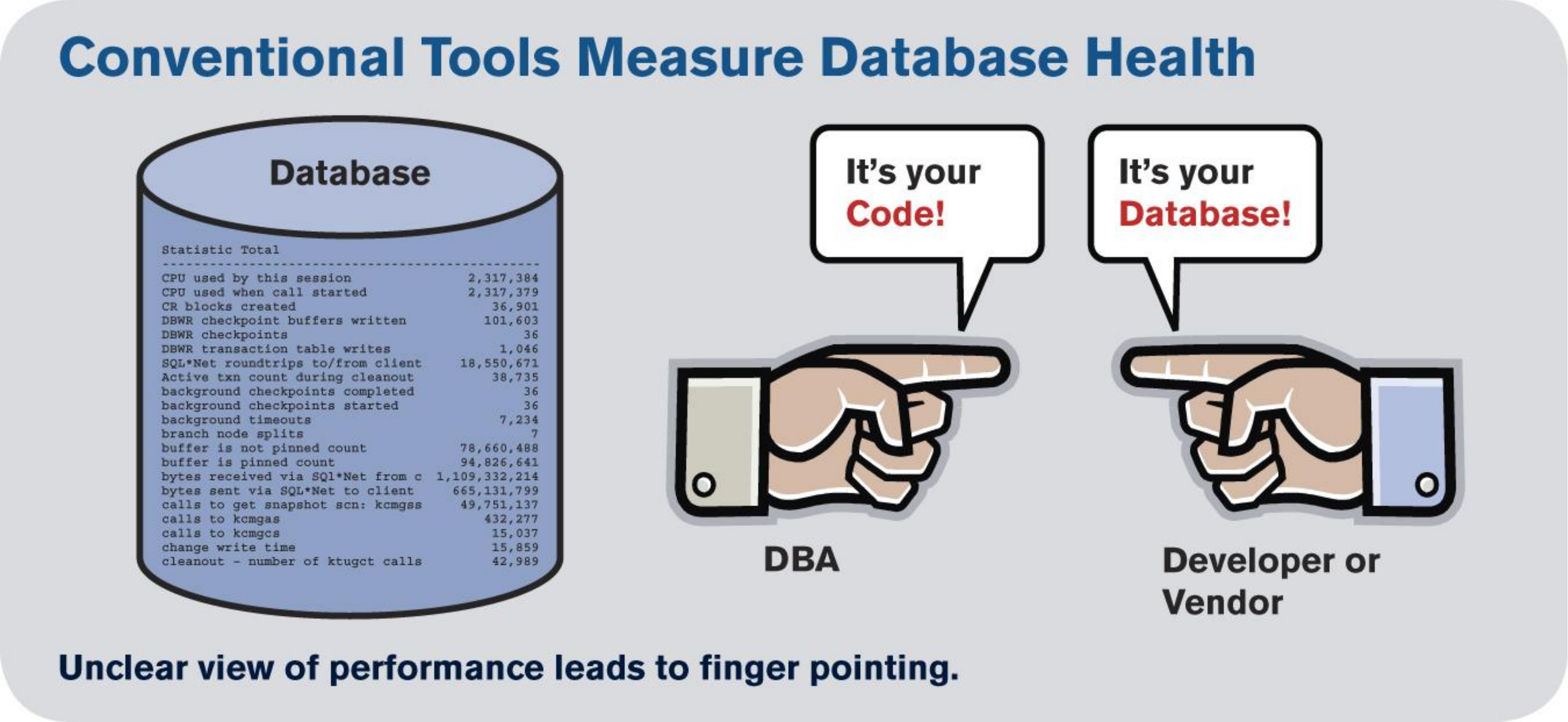

Why Wait Time Analysis Rocks

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- Allows you to focus on *the* problem
  - Four possible resource bottlenecks

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- Waits help you to answer the BIG question:
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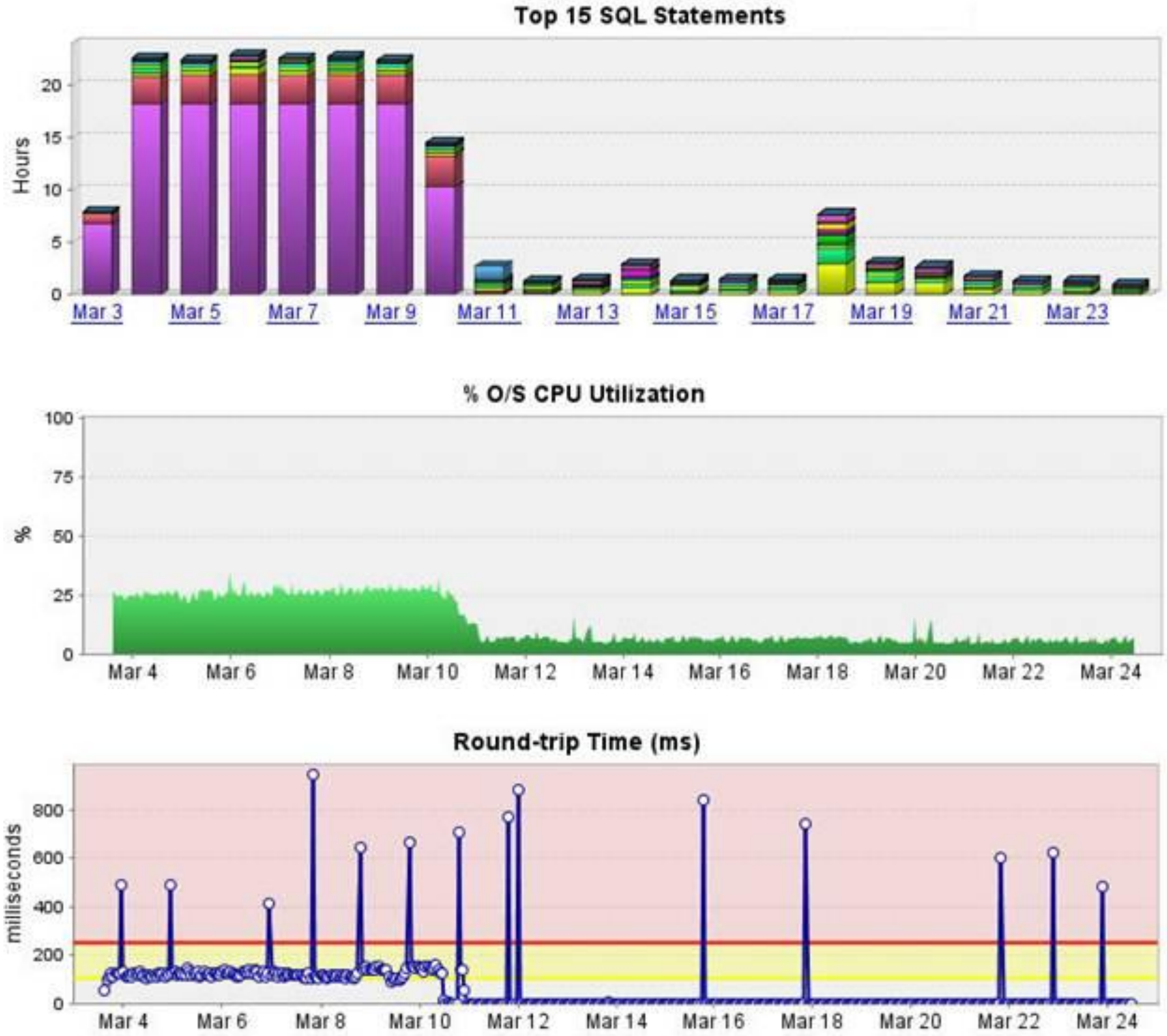
- Does not rely on Health Stats alone
  - Perfmon counters only?

Use Case (or, My Story)



How To Measure?

- 3rd party tools
- Native tools
  - DMVs
  - Perfmon
  - xEvents
  - MDW/UCP

Know What You Are Measuring

- Know difference between “wait” and “queue”
  - Wait – Any time a session is waiting
  - Queue – A measure of system resource and utilization (think perfmon)

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  - Which one is more likely to see parallelism?
  - Which one is likely to see more blocking/locking?
- What does all that mean to the DBA?

Operating With Precision

For More Information

- http://sqlcat.com/whitepapers/archive/2007/11/19/sql-server-2005-waits-and-queues.aspx
- http://technet.microsoft.com/en-us/sqlserver/gg508897.aspx

For More Information

- http://sqlskills.com/BLOGS/PAUL/post/Wait-statistics-or-please-tell-me-where-it-hurts.aspx
- https://sqlserverperformance.wordpress.com/
- http://thomaslarock.com/presentations
- http://speakerrate.com/speakers/3000-sqlrockstar

Thanks!

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THANK YOU DALLAS!

NEXT STOP: LOUISVILLE!

Questions?

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